

Ja-Chan Lu

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EDUCATION

University of California San Diego

Degree (B.S.), Data Science

Expected: June 2027

La Jolla, California

PROJECTS

Multi-Sector Equity Crash Study

- Longitudinal analysis in R of 39 equities across 5 sectors (Tech, Healthcare, Energy, Financial, Consumer Staples) from 2019–2023. Max drawdown calculations, crash-period segmentation across COVID-19 and 2022 bear market. In progress of analyzing Recovery Time, Crash Beta, Cross-Crash Comparison

[League of Legends — Carry Potential ML Pipeline](#)

- Built an end-to-end pipeline on 117,000+ professional match records from Oracle's Elixir. EDA, bivariate analysis, permutation hypothesis testing, and a Decision Tree classifier. Deployed as a public web report.
- Key Finding:** Early-game gold differential at 15 minutes is the single strongest predictor of match outcome — team composition matters far less than resource control

[NBA Veteran Efficiency -- SQLite Ranking Engine](#)

- Designed a multi-table SQLite database to rank 2024-25 season veterans by a custom per-minute relative efficiency score benchmarked against positional league averages across all 5 positions.
- Key Finding:** Versatility and court vision define veteran success — not athleticism. Older veterans who excel in assists-per-minute outrank higher scorers in the efficiency model.

[NBA Player Performance -- Predictive Modeling](#)

- Led a team of 5 through the full data lifecycle to analyze ~6,000 NBA player-seasons via the NBA API. Built shared pipelines and reproducible outputs to evaluate how height and 3-point rates correlated across three historical eras.
- Key Finding:** Height remained a significant negative predictor of 3-point attempt rate across all eras, but a 1,906% surge in 3-point attempts among Bigs from 1996 to 2024 outpaced Guards and Wings, marking the rise of the "stretch big" archetype.

WORK EXPERIENCE

University of California San Diego Cantonese Culture Club

January 2024 – April 2024

Event Coordinator

La Jolla, California

- Spearheaded the end-to-end planning and execution of 2+ club events, coordinating logistics, scheduling, and activities designed to celebrate and promote Cantonese culture.
- Drove member retention and club growth by actively participating in events, facilitating icebreakers, and building meaningful peer connections to create an inclusive community for new members.
- Served as a cultural ambassador, designing engaging programming that fostered cross-cultural dialogue and increased campus awareness of Cantonese traditions, language, and heritage.

Chinatown Service Center

October 2022 – June 2023

Intern Tutor

Los Angeles, California

- Tracked students of all ages' progress to identify learning gaps and adjust instructions while leading a cohort of 6th-graders with measurable improvements in test scores and participation.
- Communicated complex concepts to ESL students and families in Chinese

Downtown Magnets High School

August 2022 – June 2023

College Peer Counselor

Los Angeles, California

- Produced documentation to keep 12 other peers aligned on college application deadlines
- Managed internal catering operations by executing monthly face-to-face outreach to 20+ faculty members, tracking food orders, and ensuring seamless delivery logistics.

CERTIFICATIONS, SKILLS & INTERESTS

- Skills:** R Programming, Pandas, Numpy, scikit-learn, PyTorch, tidyverse, ggplot2, tidyquant, SQLite, Jupyter, Git, GitHub, Machine Learning, Data Visualization, Financial Analysis, Database design, Hypothesis testing, Time Series, EDA, MLOps
- Interests:** Weightlifting; Basketball; Cafe-Visiting; Piano;
- Languages:** Fluent in English, Proficient in Chinese (Cantonese/Mandarin)